

Microsoft Azure portal interface showing the 'haseebstore' storage account. The left sidebar lists navigation options: Overview, Activity log, Tags, Diagnose and solve problems, Access Control (IAM), Data migration, Events, Storage browser, Partner solutions, Resource visualizer, and Data storage. The 'Containers' section is selected, displaying a table of containers.

Name	Last modified	Anonymous access level	Lease state
\$logs	8/26/2025, 10:18:10 AM	Private	Available
capstone2	8/26/2025, 10:30:23 AM	Private	Available
raw	8/26/2025, 10:19:04 AM	Private	Available

The URL bar shows the path: <https://portal.azure.com/#@michaelcloudkraft@hotmail.onmicrosoft.com/resource/subscriptions/d008c7cc-9795-4292-bf79-74ae52cda63d/r...>

Microsoft Azure portal interface showing the 'capstone2' container. The left sidebar lists navigation options: Overview, Diagnose and solve problems, Access Control (IAM), and Settings. The 'Overview' section is selected, displaying a table of blobs.

Name	Last modified	Access tier	Blob type	Size	Lease state
products	8/26/2025, 10:30:38 AM				
sales	8/26/2025, 10:30:45 AM				

The URL bar shows the path: https://portal.azure.com/#view/Microsoft_Azure_Storage/ContainerMenuBlade/~/-/overview/storageAccountId/%2Fsubscriptions%2Fd008c7...

Home > haseebstore | Access Control (IAM)

Add role assignment

Role

Members

Conditions

Review + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

Job function roles

Privileged administrator roles

Grant access to Azure resources based on job function, such as the ability to create virtual machines.

blob

Type: All

Category: All

Name	Description	Type	Category	Details
CreateBlobStorage		CustomRole	None	View
Defender CSPM Storage Data Scan...	Grants access to read blobs and files. This role is used by the data scanner of Dfender CSPM.	BuiltInRole	None	View
Defender for Storage Data Scanner	Grants access to read blobs and update index tags. This role is used by the data scanner of...	BuiltInRole	None	View
Storage Blob Data Contributor	Allows for read, write and delete access to Azure Storage blob containers and data	BuiltInRole	Storage	View
Storage Blob Data Owner	Allows for full access to Azure Storage blob containers and data, including assigning POSI...	BuiltInRole	Storage	View
Storage Blob Data Reader	Allows for read access to Azure Storage blob containers and data	BuiltInRole	Storage	View
Storage Blob Delegator	Allows for generation of a user delegation key which can be used to sign SAS tokens	BuiltInRole	Storage	View

Review + assign

Previous

Next

Feedback

10:39 AM

8/26/2025

Home > haseebstore | Access Control (IAM)

Add role assignment

Role

Members

Conditions

Review + assign

Showing a filtered list of roles because your permissions include a condition. [Learn more](#)

[View my access](#)

Selected role

Storage Blob Data Contributor

Assign access to

User, group, or service principal

Managed identity

Members

+ Select members

Name	Object ID
No members selected	

Description

Optional

Review + assign

Previous

Next

Feedback

Select managed identities

Some results might be hidden due to your ABAC condition.

Subscription *

LabsKraft

Managed identity

Access Connector for Azure Databricks (22)

haseeb

Selected members:

haseebs_connector

/subscriptions/d008c7cc-9795-4292-bf79-74ae52cda63d/resourceGroups/...

Remove

Select

Close

Feedback

10:39 AM

8/26/2025

Microsoft Azure portal showing the configuration for the **haseebs_connector** (Access Connector for Azure Databricks).

Overview

- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Settings
- Automation
- Help

Essentials

Property	Value
Resource group	LabsKraft2027
Location	Central India
Subscription	LabsKraft
Subscription ID	d008c7cc-9795-4292-bf79-74ae52cda63d
Tags	

State: Succeeded

Resource ID: /subscriptions/d008c7cc-9795-4292-bf79-74ae52cda63d/resourceGroups/...

Databricks portal showing the configuration for a new external location.

Create a new external location

An external location allows you to access your data stored in cloud storage (e.g. Azure Data Lake Storage). You will need the cloud storage path and a paired credential (e.g. managed identity) which gives access to that path. [Learn more](#)

External location name*

haseeb_cap2

Storage type*

Azure Data Lake Storage

URL*

Enter the bucket path that you want to use as the external location. Note: This must be an ADLS Gen2 storage account with a hierarchical namespace

abfss://capstone2@haseebstore.dfs.core.windows.net/ [Copy from DBFS](#)

Storage credential* [Learn more](#)

Provide a storage credential capable of accessing the URL

+ Create new storage credential

Access connector ID [Learn more](#)

2-bf79-74ae52cda63d/resourceGroups/LabsKraft2027/providers/Microsoft.Databricks/accessConnectors/haseebs_connector

User assigned managed identity ID (optional)

Comment

Microsoft Azure databricks

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Serverless Starter Warehouse

Create a new catalog

A catalog is the first layer of Unity Catalog's three-level namespace and is used to organize your data assets. [Learn more](#)

Catalog name*

haseebcapstone2

Type*

Standard

Storage location

haseebcap sub/path

Create a new external location

abfss://capstone2@haseebstore.dfs.core.windows.net/

Location in cloud storage where data for managed tables will be stored. If not specified, the location will default to the metastore root location.

Cancel Create

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Playground

Experiments

Features

Delta Sharing

Clean Rooms

External Data

Add data

Send feedback

Filter

Last viewed	Type
6 minutes ago	View
7 minutes ago	Catalog
7 minutes ago	Catalog
19 hours ago	Schema
19 hours ago	Streaming table
19 hours ago	Schema
19 hours ago	Catalog
22 hours ago	Schema

10:41 AM 8/26/2025

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Serverless Starter Warehouse Serverless S

Quick access

Recents Favorites Catalogs

Name	Last
catalog_privileges	
haseebcapstone2.information_schema	7 mi
haseebcapstone2	7 mi
haseebcat	8 mi
capstone_retail_hetul_haseebcat	19 hours ago
bronze_orders_haseebcat.capstone_retail_haseeb	19 hours ago
capstone_retail_haseeb_haseebcat	19 hours ago
haseebcatalog	19 hours ago
retail_demo_haseebcatalog	22 hours ago

Schema

Streaming table

Schema

Catalog

Schema

Settings

Azure portal

Privacy policy

Previews

Send feedback

Log out

10:42 AM 8/26/2025

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Workspace admin

Appearance

Identity and access

Security

Compute

Development

Notifications

Advanced

User

Profile

Preferences

Developer

Linked accounts

Notifications

Identity and access

Management and permissions

Users
Manage users and entitlements [Manage](#)

Groups
Manage groups and entitlements [Manage](#)

Service principals
Manage service principals and entitlements [Manage](#)

10:42 AM 8/26/2025

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Workspace settings > Identity and access >

Users

Filter users 15 total [Add user](#)

Add user

Enter the email address of your new user.

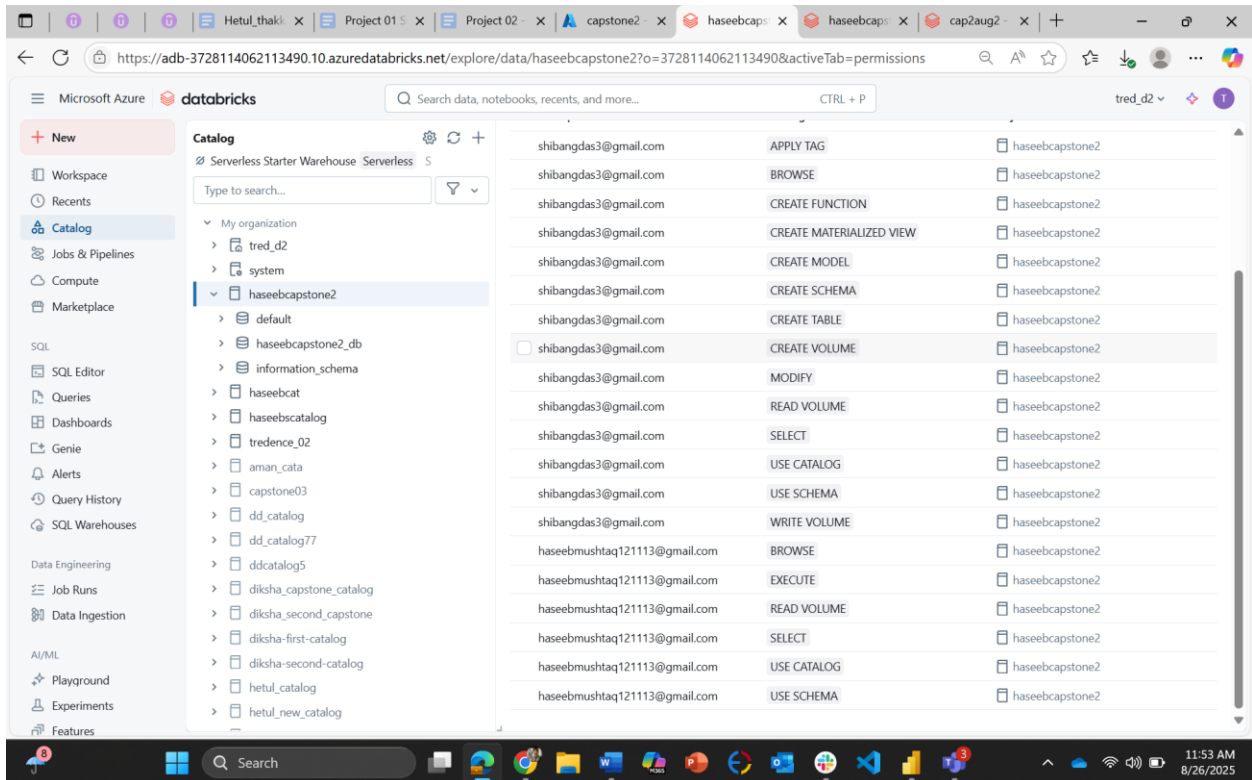
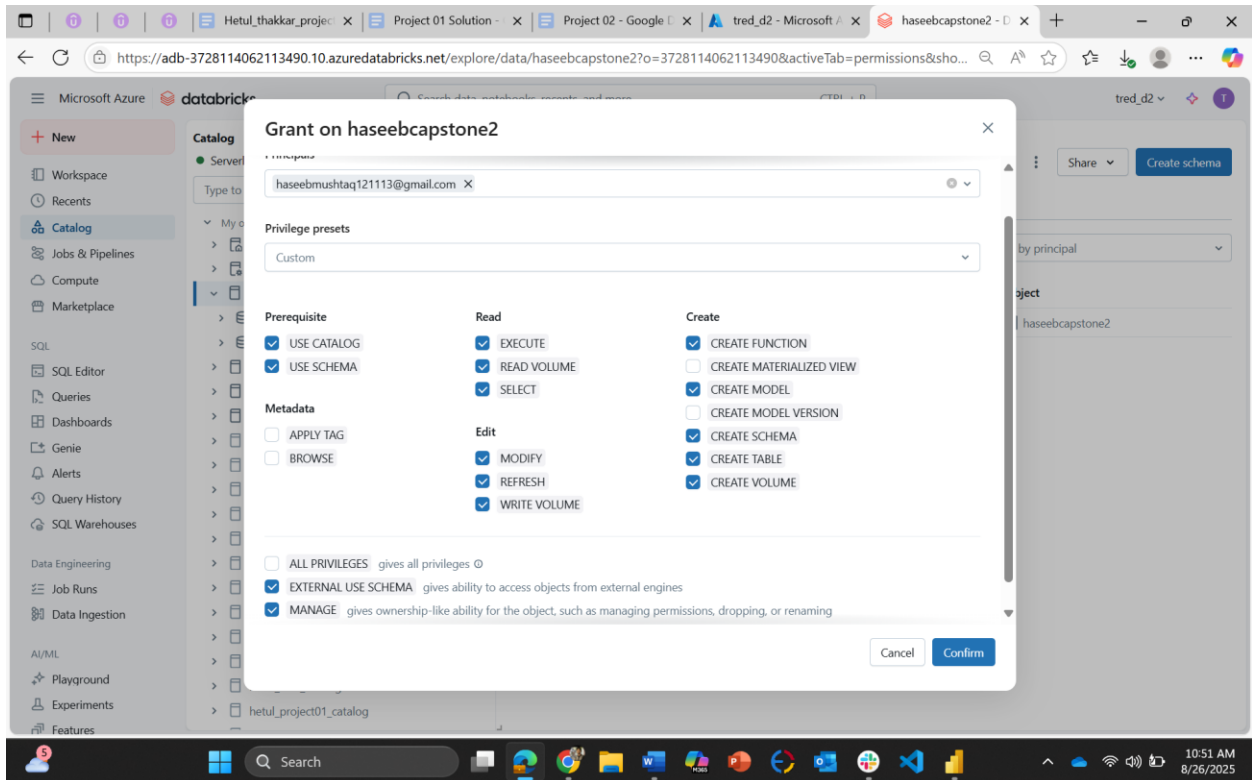
New user email

haseebmushtaq121113@gmail.com

[Back](#) [Add](#)

✓	michael.kloudkraft_hotmail.com#ext#@michaelkloudkrafthotmail.onmicrosoft.com	Michael A
✓	pimeyi1489@ahanim.com	
✓	rohit@michaelkloudkrafthotmail.onmicrosoft.com	RohitY
✓	tredence12@michaelkloudkrafthotmail.onmicrosoft.com	Tredence LabsKraft12
✓	tredence22@michaelkloudkrafthotmail.onmicrosoft.com	Tredence LabsKraft22
✓	tredence5@michaelkloudkrafthotmail.onmicrosoft.com	Tredence LabsKraft5
✓	tredence6@michaelkloudkrafthotmail.onmicrosoft.com	Tredence LabsKraft6

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Jobs & Pipelines

haseebcapstone2 ☆ Send feedback

Run: 8/26/2025, 11:50:22 AM · Completed Select tables for refresh

Graph List

Pipeline details Update details

Pipeline ID 6cd01799-486f-497e-86f6-df974c919b07

Pipeline type ETL pipeline

Source code /Users/tredence7@microsoft.com/cap2aug2

Run as Tredence LabsKraft7

Tags None

Event log Query history

All Info Warning Error Filter...

3 minutes ago flow_progress Flow 'haseebcapstone2.haseebcapstone2_db.gold_sales_summary' is RUNNING.

3 minutes ago dataset_life_cycle Materialized View 'haseebcapstone2.haseebcapstone2_db.gold_sales_summary' has been created.

3 minutes ago flow_progress Flow 'haseebcapstone2.haseebcapstone2_db.gold_sales_summary' has COMPLETED.

3 minutes ago update_progress Update d8038e is COMPLETED.

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cap2aug2

File Edit View Run Help Python Tabs: ON ☆ Last edit was 5 minutes ago

Last execution failed 1

```
1 import dlt
2 from pyspark.sql.functions import col, to_date, sum, countDistinct
3 from pyspark.sql.types import StructType, StructField, StringType, IntegerType, DoubleType
4
5 # SALES SCHEMA
6 sales_schema = StructType([
7     StructField("sale_id", StringType(), True),
8     StructField("product_id", StringType(), True),
9     StructField("region", StringType(), True),
10    StructField("store_id", StringType(), True),
11    StructField("quantity", IntegerType(), True),
12    StructField("sales_amount", DoubleType(), True),
13    StructField("sale_date", StringType(), True)
14 ])
15
16 # PRODUCTS SCHEMA (common structure for both csv and JSON files)
17 products_schema = StructType([
18     StructField("product_id", StringType(), True),
19     StructField("product_name", StringType(), True),
20     StructField("category", StringType(), True),
21     StructField("price", DoubleType(), True),
22     StructField("discount_rate", DoubleType(), True)
23 ])
24
25 # BRONZE LAYER
26
27
```

This notebook is associated with 1 ETL pipeline

To attach to a pipeline, click the compute selector in the notebook toolbar and select a pipeline from the dropdown menu.

Do not show this again

Notebook detached

Cluster 0821-050901-erbo36or not usable

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cap2aug2

File Edit View Run Help Python Tabs: ON Last edit was 5 minutes ago

1 Last execution failed

```
29 @dlt.table(name="bronze_sales")
30 def bronze_sales():
31     return (
32         spark.readStream.format("cloudFiles")
33         .option("cloudFiles.format", "csv")
34         .schema(sales_schema)
35         .load("abfss://capstone2@haseebstore.dfs.core.windows.net/sales/")
36     )
37
38 # Read Products CSV
39 @dlt.table(name="bronze_products_csv")
40 def bronze_products_csv():
41     return (
42         spark.readStream.format("cloudFiles")
43         .option("cloudFiles.format", "csv")
44         .schema(products_schema)
45         .load("abfss://capstone2@haseebstore.dfs.core.windows.net/products/*.csv")
46     )
47
48 # Read Products JSON
49 @dlt.table(name="bronze_products_json")
50 def bronze_products_json():
51     return (
52         spark.readStream.format("cloudFiles")
53         .option("cloudFiles.format", "json")
54         .schema(products_schema) # explicitly provide schema (fixes error)
55         .load("abfss://capstone2@haseebstore.dfs.core.windows.net/products/*.json")
56     )
57
```

This notebook is associated with 1 ETL pipeline

To attach to a pipeline, click the compute selector in the notebook toolbar and select a pipeline from the dropdown menu.

☐ Do not show this again

Notebook detached

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cap2aug2

File Edit View Run Help Python Tabs: ON Last edit was 5 minutes ago

1 Last execution failed

```
59 # SILVER LAYER (Cleaned)
60 # -----
61
62 @dlt.table(name="silver_sales")
63 @dlt.expect("Valid sales amount", "sales_amount >= 0")
64 @dlt.expect("Valid quantity", "quantity > 0")
65 def silver_sales():
66     df = dlt.read("bronze_sales")
67     return (
68         df.dropna()
69         .withColumn("sales_amount", col("sales_amount").cast("double"))
70         .withColumn("quantity", col("quantity").cast("int"))
71         .withColumn("sale_date", to_date("sale_date", "yyyy-MM-dd"))
72     )
73
74 # Union both product sources
75 @dlt.table(name="silver_products")
76 def silver_products():
77     csv_df = dlt.read("bronze_products_csv")
78     json_df = dlt.read("bronze_products_json")
79     return (
80         csv_df.unionByName(json_df)
81         .dropna()
82         .dropDuplicates(["product_id"])
83     )
84
85 # -----
86 # GOLD LAYER
87 # -----
```

This notebook is associated with 1 ETL pipeline

To attach to a pipeline, click the compute selector in the notebook toolbar and select a pipeline from the dropdown menu.

☐ Do not show this again

Notebook detached

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cap2aug2 x +

File Edit View Run Help Python Tabs: ON Last edit was 1 minute ago

Notebook detached
Cluster 0821-050901-erbo36or not usable

```
1 import dlt
2 from pyspark.sql.functions import *
3
4 # Ingest raw sales data (csv files)
5 @dlt.table(
6     name="sales_raw",
7     comment="Raw sales data ingested from CSV files",
8     table_properties={"quality": "bronze"}
9 )
10 def sales_raw():
11     return (
12         spark.readStream.format("cloudFiles") # Auto loader for incremental loads
13         .option("cloudFiles.format", "csv")
14         .option("header", "true")
15         .load("abfss://capstone2@haseebstore.dfs.core.windows.net/sales/")
16     )
17
18 # Ingest raw product data (CSV + JSON)
19 @dlt.table(
20     name="products_raw",
21     comment="Raw product data ingested from CSV and JSON files",
22     table_properties={"quality": "bronze"}
23 )
24 def products_raw():
25     csv_df = (
26         spark.readStream.format("cloudFiles")
27         .option("cloudFiles.format", "csv")
28         .option("header", "true")
29         .load("abfss://capstone2@haseebstore.dfs.core.windows.net/products/")
30     )
31     json_df = (
32         spark.readStream.format("cloudFiles")
33         .option("cloudFiles.format", "json")
34         .load("abfss://capstone2@haseebstore.dfs.core.windows.net/products/")
35     )
36     return csv_df.unionByName(json_df, allowMissingColumns=True)
37
38 # Clean sales data
39 @dlt.table(
40     name="sales_clean",
41     comment="Cleaned sales data with enforced schema",
42     table_properties={"quality": "silver"}
43 )
44 @dlt.expect("valid_sales_amount", "sales_amount >= 0")
45 @dlt.expect("valid_quantity", "quantity > 0")
46 def sales_clean():
47     return (
48         dlt.read("sales_raw")
49         .withColumn("sale_id", col("sale_id").cast("string"))
50         .withColumn("product_id", col("product_id").cast("string"))
51         .withColumn("region", col("region").cast("string"))
52         .withColumn("store_id", col("store_id").cast("string"))
53         .withColumn("quantity", col("quantity").cast("int"))
```

2:25 PM 8/26/2025

Microsoft Azure databricks

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cap2aug2 x +

File Edit View Run Help Python Tabs: ON Last edit was 1 minute ago

Notebook detached
Cluster 0821-050901-erbo36or not usable

```
25 csv_df = (
26     spark.readStream.format("cloudFiles")
27     .option("cloudFiles.format", "csv")
28     .option("header", "true")
29     .load("abfss://capstone2@haseebstore.dfs.core.windows.net/products/")
30 )
31 json_df = (
32     spark.readStream.format("cloudFiles")
33     .option("cloudFiles.format", "json")
34     .load("abfss://capstone2@haseebstore.dfs.core.windows.net/products/")
35 )
36 return csv_df.unionByName(json_df, allowMissingColumns=True)
37
38 # Clean sales data
39 @dlt.table(
40     name="sales_clean",
41     comment="Cleaned sales data with enforced schema",
42     table_properties={"quality": "silver"}
43 )
44 @dlt.expect("valid_sales_amount", "sales_amount >= 0")
45 @dlt.expect("valid_quantity", "quantity > 0")
46 def sales_clean():
47     return (
48         dlt.read("sales_raw")
49         .withColumn("sale_id", col("sale_id").cast("string"))
50         .withColumn("product_id", col("product_id").cast("string"))
51         .withColumn("region", col("region").cast("string"))
52         .withColumn("store_id", col("store_id").cast("string"))
53         .withColumn("quantity", col("quantity").cast("int"))
```

2:25 PM 8/26/2025

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cap2aug2 x +

File Edit View Run Help Python Tabs: ON Last edit was 2 minutes ago

1

Python

Notebook detached
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```
54 .withColumn("sales_amount", col("sales_amount").cast("double"))
55 .withColumn("sale_date", to_date(col("sale_date"), "yyyy-MM-dd"))
56 .na.drop()
57 )
58
59 # Clean products data
60 @dlt.table(
61     name="products_clean",
62     comment="Cleaned products data with schema evolution support",
63     table_properties={"quality": "silver"}
64 )
65 def products_clean():
66     return (
67         dlt.read("products_raw")
68         .withColumn("product_id", col("product_id").cast("string"))
69         .withColumn("product_name", col("product_name").cast("string"))
70         .withColumn("category", col("category").cast("string"))
71         .withColumn("price", col("price").cast("double"))
72         .withColumn("discount_rate", col("discount_rate").cast("double"))
73     )
74
75 # Join sales + products and calculate KPIs
76 @dlt.table(
77     name="sales_summary",
78     comment="Daily sales summary by product and region",
79     table_properties={"quality": "gold"}
80 )
81 def sales_summary():
82     sales = dlt.read("sales_clean")
```

2:26 PM
8/26/2025

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cap2aug2 x +

File Edit View Run Help Python Tabs: ON Last edit was 2 minutes ago

1

Python

Notebook detached
Cluster 0821-050901-erbo36or not usable

```
73 )
74
75 # Join sales + products and calculate KPIs
76 @dlt.table(
77     name="sales_summary",
78     comment="Daily sales summary by product and region",
79     table_properties={"quality": "gold"}
80 )
81 def sales_summary():
82     sales = dlt.read("sales_clean")
83     products = dlt.read("products_clean")
84     return (
85         sales.join(products, "product_id", "left")
86         .groupBy("region", "sale_date", "product_id", "product_name", "category")
87         .agg(
88             sum("sales_amount").alias("daily_revenue"),
89             sum("quantity").alias("units_sold"),
90             avg("price").alias("avg_price"),
91             avg("discount_rate").alias("avg_discount")
92         )
93     )
94
```

The Delta Live Tables (DLT) module is not supported on this cluster. You should either [create a new pipeline](#) or use an existing pipeline to run DLT code.

> DLTImportException: Delta Live Tables module is not supported on Spark Connect clusters.

Diagnose error Debug

Assistant Quick Fix: ON

2:26 PM
8/26/2025

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File Edit View Run Help Python Tabs: ON ☆ Last edit was 5 minutes ago

86 # GOLD LAYER
87
88
89 @dlt.table(name="gold_sales_summary")
90 def gold_sales_summary():
91 sales = dlt.read("silver_sales")
92 products = dlt.read("silver_products")
93
94 joined_df = sales.join(products, on="product_id", how="inner")
95
96
97 result = (
98 joined_df
99 .withColumn("effective_price", col("price") * (1 - col("discount_rate")))
100 .withColumn("discount_amount", col("price") * col("discount_rate") * col("quantity"))
101 .groupBy("sale_date", "region", "product_id", "product_name", "category")
102 .agg(
103 sum("quantity").alias("total_units_sold"),
104 sum("sales_amount").alias("total_revenue"),
105 sum("discount_amount").alias("total_discount_given"),
106 sum("effective_price").alias("total_effective_price")
107)
108)
109
110 return result

1 Last execution failed

This notebook is associated with 1 ETL pipeline

To attach to a pipeline, click the compute selector in the notebook toolbar and select a pipeline from the dropdown menu.

☐ Do not show this again

Notebook detached

Cluster 0821-050901-erbo36or not usable

The Delta Live Tables (DLT) module is not supported on this cluster. You should either [create a new pipeline](#) or use an existing pipeline to run DLT code.

11:55 AM
8/26/2025

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Jobs & Pipelines

hasee

Run: 312126

Pipeline event log details

Summary JSON

ERROR 2025-08-26 14:34:55 IST update_progress

Update 312126 has failed while trying to update table 'haseebcapstone2.haseebcapstone2_db.sales_raw'. [Diagnose error](#)

Error details

`com.databricks.pipelines.common.errors.DLTAnalysisException: [CANNOT_UPDATE_TABLE_SCHEMA] Failed to merge the current and new schemas for table haseebcapstone2.haseebcapstone2_db.__materialization_mat_733ba4a4_a414_494c_ab5c_dced45e7997e_sales_raw_1.`

To proceed with this schema change, you can trigger a full refresh of this table.
Depending on your use case and the schema changes, you may be able to obviate the schema change -- you can update your queries so the output schema is compatible with the existing schema (e.g, by explicitly casting columns to the correct data type).

`com.databricks.sql.transaction.tahoe.DeltaAnalysisException: [DELTA_FAILED_TO_MERGE_FIELDS] Failed to merge file ids 'quantity' and 'quantity'`

`com.databricks.sql.transaction.tahoe.DeltaAnalysisException: [DELTA_MERGE_INCOMPATIBLE_DATATYPE] Failed to merge incompatible data types StringType and IntegerType`

Event log

All

1 minute ago dataset_life_cycle STREAMING_TABLE 'haseebcapstone2.haseebcapstone2_db.products_raw_json' has been created.

1 minute ago dataset_life_cycle STREAMING_TABLE 'haseebcapstone2.haseebcapstone2_db.products_raw_csv' has been created.

1 minute ago update_progress Update 312126 has failed while trying to update table 'haseebcapstone2.haseebcapstone2_db.sales_raw'.

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Jobs & Pipelines

hasee

Run: e9f30c

Pipeline event log details

Summary JSON

ERROR 2025-08-26 14:44:30 IST update_progress

Update e9f30c has failed while trying to update table 'haseebcapstone2.haseebcapstone2_db.sales_raw'. [Diagnose error](#)

Error details

`com.databricks.pipelines.common.errors.DLTAnalysisException: [CANNOT_UPDATE_TABLE_SCHEMA] Failed to merge the current and new schemas for table haseebcapstone2.haseebcapstone2_db.__materialization_mat_733ba4a4_a414_494c_ab5c_dced45e7997e_sales_raw_1.`

To proceed with this schema change, you can trigger a full refresh of this table.
Depending on your use case and the schema changes, you may be able to obviate the schema change -- you can update your queries so the output schema is compatible with the existing schema (e.g, by explicitly casting columns to the correct data type).

`com.databricks.sql.transaction.tahoe.DeltaAnalysisException: [DELTA_FAILED_TO_MERGE_FIELDS] Failed to merge file ids 'quantity' and 'quantity'`

`com.databricks.sql.transaction.tahoe.DeltaAnalysisException: [DELTA_MERGE_INCOMPATIBLE_DATATYPE] Failed to merge incompatible data types StringType and IntegerType`

Event log

All

2 minutes ago update_progress Update e9f30c is INITIALIZING.

1 minute ago update_progress Update e9f30c is SETTING_UP_TABLES.

1 minute ago update_progress Update e9f30c has failed while trying to update table 'haseebcapstone2.haseebcapstone2_db.sales_raw'.

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cap2aug2 x +

File Edit View Run Help Python Tabs: ON Last edit was 2 minutes ago

Notebook detached
Cluster 0821-050901-erbo36or not usable

1 Last execution failed

```
1 import dlt
2 from pyspark.sql.functions import *
3 from pyspark.sql.types import StructType, StructField, StringType, DoubleType, IntegerType
4
5 # Define strict schemas
6 # -----
7
8 sales_schema = StructType([
9     StructField("sale_id", StringType(), True),
10    StructField("product_id", StringType(), True),
11    StructField("region", StringType(), True),
12    StructField("store_id", StringType(), True),
13    StructField("quantity", IntegerType(), True),
14    StructField("sales_amount", DoubleType(), True),
15    StructField("sale_date", StringType(), True)
16 ])
17
18 products_schema = StructType([
19     StructField("product_id", StringType(), True),
20     StructField("product_name", StringType(), True),
21     StructField("category", StringType(), True),
22     StructField("price", DoubleType(), True),
23     StructField("discount_rate", DoubleType(), True)
24 ])
25
26 # -----
27 # Ingest raw sales data (drops malformed rows)
28 # dlt + shifter
```

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cap2aug2 x +

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1 Last execution failed

```
28 # -----
29 @dlt.table(
30     name="sales_raw",
31     comment="Raw sales data ingested from CSV files",
32     table_properties={"quality": "bronze"}
33 )
34 def sales_raw():
35     return (
36         spark.readStream.format("cloudFiles")
37         .option("cloudFiles.format", "csv")
38         .option("header", "true")
39         .option("mode", "DROPMALFORMED") # Automatically drop malformed rows
40         .schema(sales_schema)
41         .load("abfss://capstone2@haseebstore.dfs.core.windows.net/sales/")
42     )
43
44 # -----
45 # Ingest raw products data (CSV + JSON) with malformed row drop
46 # -----
47 @dlt.table(
48     name="products_raw_csv",
49     comment="Raw product data from CSV",
50     table_properties={"quality": "bronze"}
51 )
52 def products_raw_csv():
53     return (
54         spark.readStream.format("cloudFiles")
55         .option("cloudFiles.format", "csv")
56         .option("header", "true")
57         .load("abfss://capstone2@haseebstore.dfs.core.windows.net/products/")
58     )
```

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Python

```
56 .option("header", "true")
57 .option("mode", "DROPPALFORMED")
58 .schema(products_schema)
59 .load("abfss://capstone2@haseebstore.dfs.core.windows.net/products/")
60 )
61
62 @dlt.table(
63     name="products_raw_json",
64     comment="Raw product data from JSON",
65     table_properties={"quality": "bronze"}
66 )
67 def products_raw_json():
68     return (
69         spark.readStream.format("cloudFiles")
70         .option("cloudFiles.format", "json")
71         .option("mode", "DROPPALFORMED")
72         .schema(products_schema)
73         .load("abfss://capstone2@haseebstore.dfs.core.windows.net/products/")
74     )
75
76 # -----
77 # Unified Products
78 # -----
79 @dlt.view(name="products_raw")
80 def products_raw():
81     return dlt.read("products_raw_csv").unionByName(dlt.read("products_raw_json"))
82
83 # -----
84 # Clean sales data
```

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Python

```
86 @dlt.table(
87     name="sales_clean",
88     comment="Cleaned sales data with enforced schema",
89     table_properties={"quality": "silver"}
90 )
91 @dlt.expect("valid_sales_amount", "sales_amount >= 0")
92 @dlt.expect("valid_quantity", "quantity > 0")
93 def sales_clean():
94     return (
95         dlt.read("sales_raw")
96         .withColumn("sale_date", to_date(col("sale_date"), "yyyy-MM-dd"))
97         .na.drop()
98     )
99
100 # -----
101 # Clean products data
102 # -----
103 @dlt.table(
104     name="products_clean",
105     comment="Cleaned products data",
106     table_properties={"quality": "silver"}
107 )
108 def products_clean():
109     return dlt.read("products_raw")
110
111 # -----
112 # Sales summary
113 # -----
114 @dlt.table(
```

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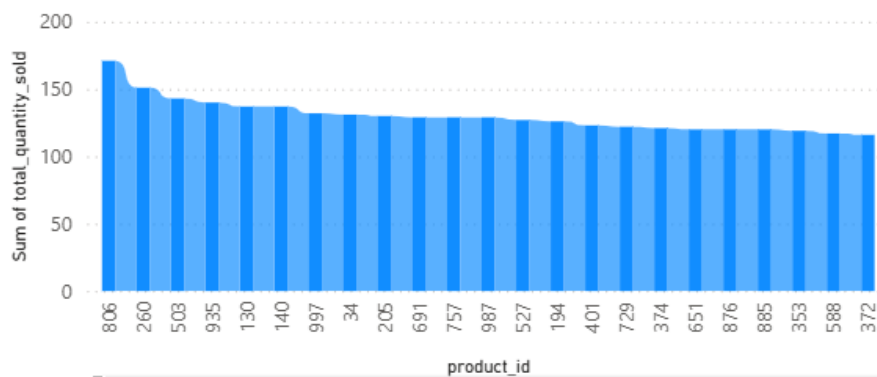
File Edit View Run Help Python Tabs: ON Last edit was 3 minutes ago

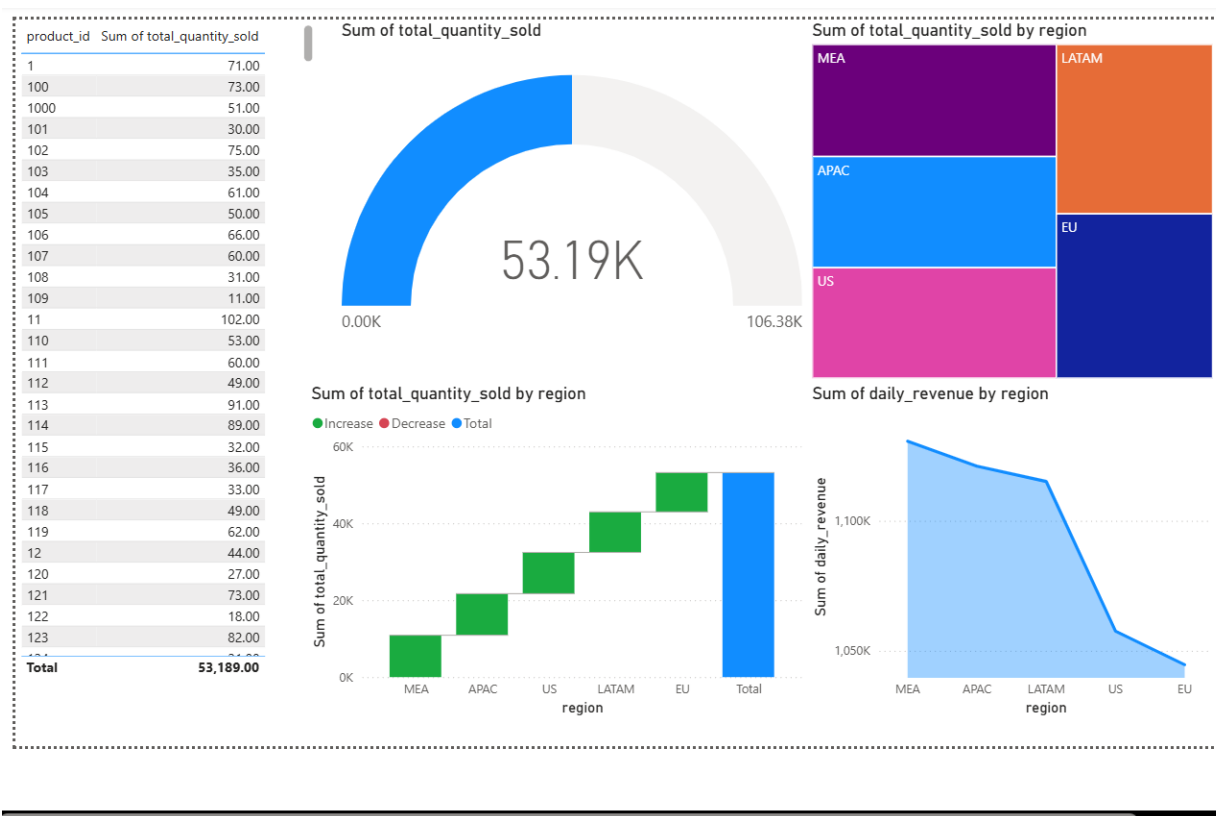
Notebook detached
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```
104 name="products_clean",
105 comment="Cleaned products data",
106 table_properties={"quality": "silver"}
107 )
108 def products_clean():
109     return dlt.read("products_raw")
110
111 # -----
112 # Sales summary
113 # -----
114 @dlt.table(
115     name="sales_summary",
116     comment="Daily sales summary by product and region",
117     table_properties={"quality": "gold"}
118 )
119 def sales_summary():
120     sales = dlt.read("sales_clean")
121     products = dlt.read("products_clean")
122     return (
123         sales.join(products, "product_id", "left")
124         .groupBy("region", "sale_date", "product_id", "product_name", "category")
125         .agg(
126             sum("sales_amount").alias("daily_revenue"),
127             sum("quantity").alias("units_sold"),
128             avg("price").alias("avg_price"),
129             avg("discount_rate").alias("avg_discount")
130         )
131     )
132
```

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Sum of total_quantity_sold by product_id





```
import dlt
```

```
from pyspark.sql.functions import col, to_date, sum, countDistinct
```

```
from pyspark.sql.types import StructType, StructField, StringType, IntegerType, DoubleType,
DateType
```

```
# SALES SCHEMA
```

```
sales_schema = StructType([
    StructField("sale_id", StringType(), True),
    StructField("product_id", StringType(), True),
    StructField("region", StringType(), True),
    StructField("store_id", StringType(), True),
    StructField("quantity", IntegerType(), True),
    StructField("sales_amount", DoubleType(), True),
    StructField("sale_date", StringType(), True)
```

```
])
```

```

# PRODUCTS SCHEMA (common structure for both CSV and JSON files)

products_schema = StructType([
    StructField("product_id", StringType(), True),
    StructField("product_name", StringType(), True),
    StructField("category", StringType(), True),
    StructField("price", DoubleType(), True),
    StructField("discount_rate", DoubleType(), True)
])

# -----

# BRONZE LAYER

# -----

@dlt.table(name="bronze_sales")
def bronze_sales():
    return (
        spark.readStream.format("cloudFiles")
        .option("cloudFiles.format", "csv")
        .schema(sales_schema)
        .load("abfss://capstone2@haseebstore.dfs.core.windows.net/sales/")
    )

# Read Products CSV

@dlt.table(name="bronze_products_csv")
def bronze_products_csv():
    return (
        spark.readStream.format("cloudFiles")
        .option("cloudFiles.format", "csv")
        .schema(products_schema)

```

```

        .load("abfss://capstone2@haseebstore.dfs.core.windows.net/products/*.csv")
    )

# Read Products JSON
@dlt.table(name="bronze_products_json")
def bronze_products_json():
    return (
        spark.readStream.format("cloudFiles")
        .option("cloudFiles.format", "json")
        .schema(products_schema) # explicitly provide schema (fixes error)
        .load("abfss://capstone2@haseebstore.dfs.core.windows.net/products/*.json")
    )

# -----
# SILVER LAYER (Cleaned)
# -----

@dlt.table(name="silver_sales")
@dlt.expect("Valid sales amount", "sales_amount >= 0")
@dlt.expect("Valid quantity", "quantity > 0")
def silver_sales():
    df = dlt.read("bronze_sales")
    return (
        df.dropna()
        .withColumn("sales_amount", col("sales_amount").cast("double"))
        .withColumn("quantity", col("quantity").cast("int"))
        .withColumn("sale_date", to_date("sale_date", "yyyy-MM-dd"))
    )

# Union both product sources

```



```

@dlt.table(name="silver_products")

def silver_products():

    csv_df = dlt.read("bronze_products_csv")

    json_df = dlt.read("bronze_products_json")

    return (

        csv_df.unionByName(json_df)

        .dropna()

        .dropDuplicates(["product_id"])

    )

# -----

# GOLD LAYER

# -----

@dlt.table(name="gold_sales_summary")

def gold_sales_summary():

    sales = dlt.read("silver_sales")

    products = dlt.read("silver_products")

    joined_df = sales.join(products, on="product_id", how="inner")

    result = (

        joined_df

        .withColumn("effective_price", col("price") * (1 - col("discount_rate")))

        .withColumn("discount_amount", col("price") * col("discount_rate") * col("quantity"))

        .groupBy("sale_date", "region", "product_id", "product_name", "category")

        .agg(

            sum("quantity").alias("total_units_sold"),

            sum("sales_amount").alias("total_revenue"),

            sum("discount_amount").alias("total_discount_given"),

```

```
        sum("effective_price").alias("total_effective_price")
    )
)
return result
```